# Sunbeam Infotech

Exploring new ideas, Reaching new heights!

# STL

- STL is part of C++ standard.
- It has template implementations of common data structures.
- STL has three main components
  - Containers
  - Algorithms
  - Iterators
- Additionally STL also have
  - Function objects
  - Allocators
  - Utility

# STL

- Containers hold data and operations to be performed on data.
- STL containers are of three types
  - Sequential: Linear collection
    - vector, list, deque
  - Associative: Key-value pair collection
    - set, map, multimap
  - Adapters: Limited container functionality
    - stack, queue

# STL

- Containers are traversed using iterators.
- Usually iterators are implemented as nested classes in containers.
- Iterators are smart pointers (with -> and * operators overloaded).
- There are six types of iterators
  - Input iterator (read ops, fwd)
  - Output iterator (write ops, fwd)
  - Bi-directional iterator (rw, bi-dirn)
  - Forward iterator (rw, fwd)
  - Reverse iterator (rw, rev)
  - Random access iterator (rw, any)

# STL

- Algorithms are global functions that operates on containers.
- They can be classified as
  - Search functions
  - Sort functions
  - Manipulaion functions
  - Non-modifying functions
  - Numeric functions

Thank you!